

Multi-ancestry fine-mapping at scale

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Abstract

Background

Genome-wide association studies have allowed researchers to identify millions of single-variant associations to common disease and measurement traits. These associations carry substantial value for explaining biological processes. However, associated variants are correlated through linkage disequilibrium. An association signal at a locus therefore does not by itself identify which correlated variant influences the phenotype. Bayesian fine-mapping methods resolve this by assigning each variant at a locus a posterior inclusion probability — the probability that the variant itself is causal, conditioned on the observed association and the local LD structure. The variants at a locus are aggregated into a credible set that captures the true driver of the signal with a defined posterior probability^{e.g. SuSiE, [1](#)}. These methods were originally established for single-ancestry resolution, and already run at scale within the Open Targets Platform^{[2](#)}: release 26.06 includes 1,453,884 credible sets across 53,239 GWAS Catalog studies and 32,137 credible sets across 2,374 UKB-PPP studies, fine-mapped by PICS and SuSiE-inf.

Variants that tag the true signal through linkage disequilibrium, without driving it, show marginally different effects across populations due to differences in LD structure — for example, African-ancestry populations carry systematically smaller LD blocks than European-ancestry populations^{[3](#)}. A tag variant statistically indistinguishable from the true driver in one ancestry can be distinguishable in another. Simulation and empirical work demonstrates that, where the underlying variant is shared between ancestry groups, a trans-ethnic fine-mapping improves resolution over a single-ancestry analysis^{[4](#)}. Multi-ancestry fine-mapping methods exploit the differences in LD by integrating data from multiple populations^{[3,5,6](#)}.

The growth of large trans-ethnic biobanks producing GWAS for the same trait across ancestries (e.g. Pan-UKBB^{[7](#)}) allows multiple ancestries to be leveraged to narrow the credible set and improve fine-mapping accuracy for the underlying variants shared across ancestries. Grouping GWAS Catalog studies in release 26.06 by shared trait, 7,490 trait-sets already span two or more ancestries, with as many as five ancestries represented in a single set and one trait — body mass index — spanning 9,442 studies. We introduce a solution that allows for scalable

fine-mapping of GWAS studies across multiple ancestries, using a set of established methods and existing external LD reference panels.

Methods

We are developing a scalable multi-ancestry fine-mapping workflow that integrates GWAS summary statistics, using well-established Gentropy parquet-native datasets, with ancestry-matched linkage disequilibrium reference panels from gnomAD⁸ and Pan-UKBB⁷.

The workflow is implemented using Nextflow⁹ on top of the Open Targets Gentropy framework, and is designed to run in-house at biobank scale. Unifying this workflow with Open Targets' existing single-ancestry fine-mapping is left for future work.

The pipeline is structured as a sequence of pluggable stages. A resolver reads the Gentropy study index and, for each trait, determines which combinations of studies across ancestries are eligible for multi-ancestry fine-mapping. The pipeline routes each eligible combination to one or more applicable methods. Studies assigned to an execution route are then clumped into loci. Per-locus, per-ancestry linkage disequilibrium is retrieved from reference panels through a common access layer. Clumped loci and their matched LD are passed through a shared Gentropy input interface. The method itself is agnostic and selectable, spanning MultiSuSiE⁶ as proof of concept, with future work streamlined to integrate MESuSiE⁵, SuShiE¹⁰ and SuSiEx³. Method output is normalised back through a shared Gentropy output interface into the standard credible-set format, so that it can be reused efficiently by Spark-distributed jobs native to Open Targets' post-GWAS analysis, e.g. colocalisation.

This Nextflow multi-ancestry fine-mapping pipeline will allow Open Targets to produce more accurate credible sets at the scale set out above, by leveraging cross-population LD structure differences. Built on open-source methods and uniform access to external LD reference panels, it will support reuse by other groups working with multi-ancestry GWAS data. Its pluggable architecture will let researchers choose whichever method best fits their own baseline fine-mapping assumptions.

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